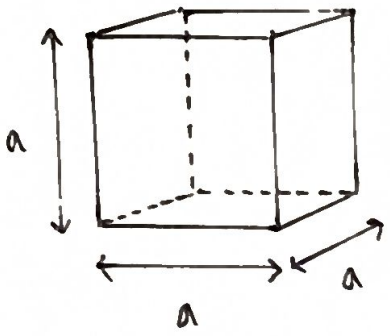
• Thermobox plant derivation



Assume internal volume of 500 mL.

Wall thickness $t = 2\,cm$.

EPS thermal conductivity $\kappa = 0,033\ W m^{-1} K^{-1}$.

$a = \sqrt[3]{5 \times 10^{-4}} = 0,0794\ m$ — $a \rightarrow$ internal side length

$A_{in} = 6a^2 = 6(0,0794)^2 = 0,0378\ m^2$ — $A_{in} \rightarrow$ internal area

$a_{out} = a + 2t = 0,0794 + 0,04 = 0,1194\ m$ — $a_{out} \rightarrow$ external side length

$A_{out} = 6a^2 = 6(0,1194)^2 = 0,0855\ m^2$ — $A_{out} \rightarrow$ external area.

$$C_{th}\frac{dT}{dt} = P_{in}(t) - \frac{T(t) - T_{amb}}{R_{th}}$$

$C_{th} \rightarrow$ thermal capacitance

$R_{th} \rightarrow$ thermal resistance

$$C_{th}\frac{d\theta}{dt} + \frac{\theta}{R_{th}} = P_{in}(t)$$

$T_{amb} \rightarrow$ ambient temperature

$P_{in} \rightarrow$ heater power.

$$L\left\{C_{th}\frac{d\theta}{dt} + \frac{\theta}{R_{th}}\right\} = L\{P_{in}(t)\}$$

Temperature rise: $\theta(t) = T(t) - T_{amb}$

$$C_{th}\,s\,\theta(s) + \frac{1}{R_{th}}\theta(s) = P_{in}(s)$$

$$\frac{\theta(s)}{P_{in}(s)} = \frac{R_{th}}{R_{th}C_{th}s + 1}$$

Duty cycle $D \in [0,1]$, therefore $P_{in} = D\frac{V_s^2}{R}$ where $P_{max} = \frac{(7,5)^2}{6,2} = 9,07\ W$

So, $G_p(s) = \frac{\theta(s)}{D(s)} = \frac{\frac{V_s^2}{R}R_{th}}{R_{th}C_{th}s+1}$

First-order prototype: $\frac{K}{\tau s + 1}$

Comparison gives: $K = \frac{V_s^2}{R}R_{th}$ and $\tau = R_{th}C_{th}$

$A_{in} = 0,0378\ m^2$, $A_{out} = 0,0855\ m^2$ $\Rightarrow$ $A_{eff} = \sqrt{A_{in}A_{out}} = \sqrt{0,0378 \times 0,0855} = 0,0568\ m^2$

$$R_{th} = \frac{t}{\kappa A_{eff}} = \frac{2 \times 10^{-2}}{0,033 \times 0,0568} = 10,66\ °C/W$$

4h

- Air. $C_{air} = \rho_{air} c_{air} V = 1,2 \times 1005 \times 5 \times 10^{-4} = 0,6$ J/°C

- Foam walls.

$V_{wall} = a^3_{out} - a^3 = (0,1194)^3 - (0,0794)^3 = 1,201 \times 10^{-3}$ m³

$\rho_{EPS} = 25$ kgm^{-3} and $c_{EPS} = 1300$ J kg^{-1}K^{-1}

$m_{wall} = 25 \times 1,201 \times 10^{-3} \approx 30$ g

$C_{wall} = 0,03 \times 1300 = 39$ J/°C

$C_{wall\ effective} = \frac{1}{2} C_{wall} = \frac{1}{2} \times 39 = 19,5$ J/°C

- Resistor. $C_{res} = 0,008 \times 800 = 6,4$ J/°C

Assume mass of 8g and $c = 800$ J kg^{-1} K^{-1}

- Total. $C_{th} = 0,6 + 19,5 + 6,4 = 26,5$ J/°C.

$K = 9,07 \times 10,66 = 96,7$

$\tau = 10,66 \times 26,5 = 283$

$$G_p(s) = \frac{96,7}{283s + 1}$$